Candidate Seat no…………

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech (EE) Examination

October 2017

EE307 Electrical Power System III

Date: 04/10/2017, Wednesday **Time: 01.30 p.m. to 04.30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1. A Why single phase representation of balanced three phase network is required for three phase calculation? **[03]**

B Explain the function of **[04]**
(i) Arc runners
(ii) Arc splitters
(iii) Magnetic blow out coil.
(iv) Arcing Contacts

Q – 2 **Attempt any TWO**

A Derive the expression for rate of rise of restriking voltage (RRRV). Find out the time for peak restriking voltage and peak of restriking voltage with necessary circuit and assumptions. **[07]**

B With neat and clean diagrams explain the phenomena of current chopping. **[07]**

C Write technical note on ionization and deionization of gases. **[07]**

Q - 3 **Answer any TWO**

A With neat diagram explain the construction and working of Air Break Circuit Breaker. **[07]**

B With neat diagram explain the construction and working of Vacuum Circuit Breaker. **[07]**

C Write technical note on rating of circuit breaker. **[07]**

SECTION – II

Q - 4 State the following statements are TRUE or FALSE. **[07]**

I. Either referred from HV side or LV side, pu impedance of the transformer remains same.
II. Neutral impedance has significance effects on Line to Line fault currents.
III. In a star connected system, without neutral grounding, zero-sequence currents are zero.
IV. The short circuit current for an unloaded synchronous machine is inversely proportional to its percentage reactance.
V. For LLG fault, one set of the possible values of sequence components is:
I_{a1}=j1.54 pu, I_{a2}= j1.54 pu and I_{a0}=j1.54 pu.

VI. A symmetrically designed generator generates voltage of positive sequence only.

VII. The power invariance theorem says that, in all sequence network power remains same.

Q – 5 Attempt Any Two

A A power station has two alternators of rating 2500 kVA & 5000 kVA and the reactance are 6% and 8% respectively. The circuit breakers are rated 150 MVA. The system is to be extended by taking supply from a grid via transformer of 5000 kVA rating having a reactance of 7.5%. If the same Circuit Breakers are to be used, find the necessary reactance to be placed between the bus-bar and the transformer to protect the switch gear. The voltage of the system at the bus-bar is 3.3 kV. (**Take base KVA= 5000 KVA**) **[07]**

B In a 3-phase 4-wire system, the currents in a, b and c lines under abnormal conditions of loading are as under. **[07]**

$I_a = 100\angle 30^\circ\ A$, $I_b = 50\angle 300^\circ\ A$, $I_c = 30\angle 180^\circ\ A$

Calculate sequence currents in a line and neutral current.

C Derive the sequences impedances of transmission line. **[07]**

Q – 6 Attempt Any Two

A A part of power system is shown in figure 1. The ratings of equipments are indicated in figure. Calculate the fault current at shown location for double line to ground fault. (**Take base as per generator 1 Rating.)** **[07]**

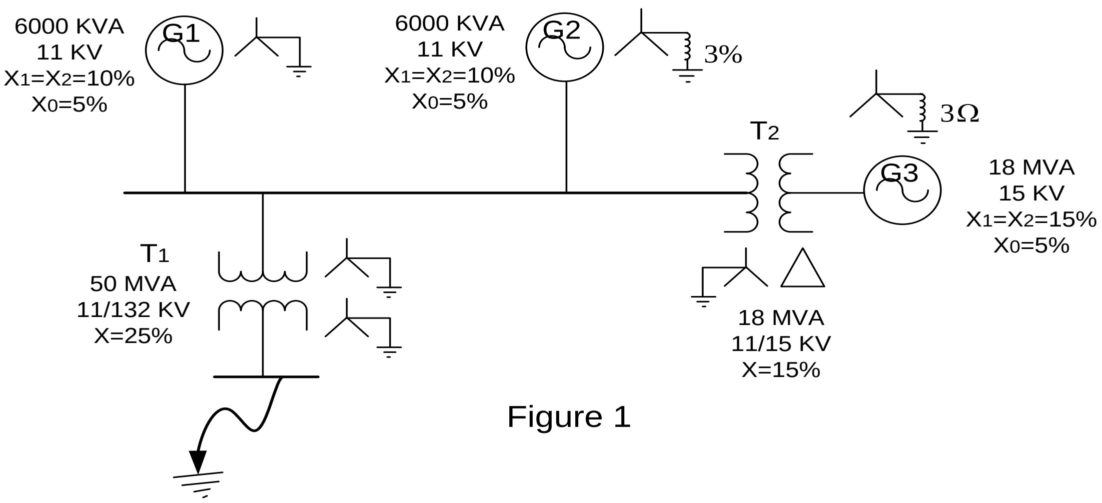

Figure 1

B Using appropriate interconnection of sequence networks, derive the equation for a line to line fault in a power system with a fault impedance of Z_f. **[07]**

C Using appropriate interconnection of sequence networks, derive the equation for a Single line to ground fault in a power system with a fault impedance of Z_f. **[07]**

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